

Xinqi Wang

Phone: (206)512-5448 Email: wangxqkaxdd@gmail.com

Personal Website: <https://elliottxinqiwan.github.io>

EDUCATION

Tsinghua University (THU), Beijing, China

Aug 2019 - July 2023

Bachelor of Engineering, Institute for Interdisciplinary Information Sciences (Yao Class)

University of Washington(UW), Seattle, WA, USA

Sep 2023 - Present

PhD in Computer Science & Engineering, Paul G. Allen School of Computer Science & Engineering

Research Interests: Machine Learning; Reinforcement Learning; Learning Theory; Deep learning; Agentic Evaluation; Automated Software Engineering

PUBLICATIONS

- [1] **X. Wang**, Q. Cui, and S. S. Du, "On Gap-dependent Bounds for Offline Reinforcement Learning," Advances in Neural Information Processing Systems, 2022.
- [2] Chuning Zhu, **Xinqi Wang**, Tyler Han, Simon S. Du, Abhishek Gupta. (2024) Transferable Reinforcement Learning via Generalized Occupancy Models. Advances in Neural Information Processing Systems, 2024.
- [3] **Xinqi Wang**, Natalia Zhang, Qiwen Cui, Runlong Zhou, Sham M. Kakade, Simon S. Du. Preference-Based Multi-Agent Reinforcement Learning: Data Coverage and Algorithmic Techniques. 2024.
- [4] **Xinqi Wang**, Hao Hu, Simon S. Du. Policy-Based Trajectory Clustering in Offline Reinforcement Learning. 2025
- [5] Liao Zhang, Tong Chen, Xiwei Wu, Qi Liu, Xiyu Zhai, **Xinqi Wang**, Qinxiang Cao. Neuro-Symbolic Generation and Validation of Memory-Aware Formal Function Specifications. 2026

RESEARCH EXPERIENCE

PBT-Bench: Evaluating AI Agents via Property-Based Testing in Large-Scale Repositories

Oct. 2025-Present

Principle researcher; Advisor: Simon S. Du

- **Conceptualized and developed PBT-Bench**, a novel evaluation framework designed to fill the gap in assessing AI agents' ability to perform document-driven debugging and property-based verification within complex software ecosystems.
- **Engineered an automated bug-injection pipeline** that systematically introduces logical and functional regressions into production-grade Python packages, requiring agents to reconcile natural language documentation with existing source code.
- **Architected the evaluation infrastructure (Infra Design)** to assess agent performance based on their ability to generate robust, property-based test functions that accurately expose injected bugs and ensure long-term code reliability.
- **Curated a diverse dataset** of test scenarios spanning high-complexity libraries, establishing a rigorous metric for "Agentic Debugging" that moves beyond simple code completion toward deep architectural understanding. Write tests for the benchmark.

Fisher-LDA Driven Benchmark Distillation for LLM Reasoning Evaluation

Apr. 2025-Sep. 2025

Principle researcher; Advisor: Tianyi Li

- **Proposed a generalized benchmark distillation framework** to optimize LLM evaluation efficiency by identifying high-discriminative power test cases across diverse reasoning modes (CoT/Think vs. Direct/No-Think).
- **Engineered a scoring metric** based on question-wise performance differentials to quantify the sensitivity of benchmarks—including AIME, GPQA, and GSM8K—to internal reasoning processes.

- **Implemented Fisher Linear Discriminant Analysis (Fisher-LDA)** to calculate importance weights for individual test items, effectively maximizing the between-class variance of reasoning behaviors while minimizing within-class noise.
- **Reduced redundant evaluation overhead** by filtering non-discriminatory problems, creating a distilled subset that maintains high alignment with full-scale benchmark results while significantly lowering inference costs.

Policy-Guided Kmeans: A Clustering Approach for Offline Reinforcement Learning Trajectories

Oct. 2024-May.2025

Principle researcher; Advisor: Simon S. Du

- Design a new algorithm, PG-Kmeans, for offline RL dataset clustering.
- Derive theoretical convergence guarantees for the algorithm.
- Design and carry out experiments to verify the superiority of PG-Kmeans over traditional VAE-based methods.

Offline MARL with Human Feedback

Feb. 2024-May.2024

Principle researcher; Advisor: Simon S. Du

- Investigate existing MARL environments and design and build the necessary MARL environment.
- Build the MARL environment database with MAPPO and VDN.
- Design and implement a code pipeline for offline multi-agent RL with human feedback.
- Design and conduct experiments to verify the role of reward regularization in reward modeling.
- Analyze the results and write the paper.

Transferable Reinforcement Learning via Generalized Occupancy Models

Sep. 2023-Feb. 2024

Principle researcher; Advisor: Simon S. Du, Abhishek Gupta

- Help design the algorithm in task-diagnostic RL settings.
- Develop performance bounds for the proposed algorithm.
- Modify experiments for verification of the algorithm.

Extra-Agent Method in Multi-agent Reinforcement Learning (MARL)

Jul. 2022-Feb. 2023

Principle researcher; Advisor: Simon S. Du

- Analyzed the inborn defects of the naive Policy Gradient (PG) method in MARL.
- Unified previous attempts and improvements based on PG into two main categories.
- Came up with a new algorithm family with performance lower bounded by PG and random search.

On Gap-dependent Bounds for Offline Reinforcement Learning

Mar. 2022-Jun. 2022

Principle researcher; Advisor: Simon S. Du, Assistant Professor in Paul G. Allen School of Computer Science & Engineering, University of Washington

- Analyzed algorithms for offline learning in tabular cases.
- Provided both the upper bounds and lower bounds of gap-dependent sample complexity.
- Provided a new technique to analyze LCB-style algorithms.
- Proved the necessity of overall optimal-policy coverage assumption for LCB.
- Work accepted by NeurIPS2022.

Causal Relationships in Autonomous Driving

Jul. 2021-Aug. 2021

Advisor: Yang Gao, Assistant Professor of Institute for Interdisciplinary Information Science, THU

- Carried out experiments over copycat problem on various auto-driving environments with different imitation learning algorithms.
- Did hacks toward environments and analyzed experiment results.

SKILLS

Programming Skills: C, Python, Pytorch, MATLAB, Latex

Language Skills:

TOEFL IBT: Total 106 (Reading 30, Listening 28, Speaking 23, Writing 25)

GRE: Total 328+3.5 (Verbal 158, Quantitative 170)

The Japanese-Language Proficiency Test, N1(The highest-level amateur examination of Japanese)